

BrickPi: Converting Raspberry Pi into a Robot

Chennai-based Dexter Industries, founded by John Cole, has come up with BrickPi—a system that converts Raspberry Pi into a robot with the help of LEGO components for crafting its working parts. We caught up with Cole to find out what exactly BrickPi is and how it works



BrickPi basic build (top view)

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When prompted what led them to design BrickPi, John Cole, founder, Dexter Industries, shares, “We had a Raspberry Pi sitting on our desks for about six months and we didn’t know what to do with it. So it was a bit of a solution looking for a problem.” Thus was born the BrickPi.

BrickPi is something that micro-robotics DIYers would definitely like to try out. As the name suggests, it is a combination of Raspberry Pi, co-opted to act as the brain of the robot, and LEGO Mindstorms sensors, bricks and motors. This project has taken the crowdfunding route—via Kickstarter—to build out a platform for playing around with robotics. It has success-

fully raised its funding goal on June 17, 2013 and received over \$125,000 in pledged backing from more than 1720 supporters—way above the original goal of \$1889.

What is a BrickPi?

BrickPi is a system that converts Raspberry Pi into a robot. The project is a board and case that connects LEGO Mindstorms sensors, motors and parts to turn your credit-card-size computer into a robot. It allows you to connect up to three NXT motors and four sensors. It has a 9V battery source that powers the motors, sensors and Raspberry Pi, un-tethering the Pi from the wall. Finally, it has a sturdy case with holes that snap fit with LEGO parts. Thus Raspberry Pi is the brain of a BrickPi joined with the LEGO building system.

The system comprises BrickPi board and the case. The BrickPi board slides over your Raspberry Pi,

connects, controls and powers Mindstorms motors and sensors, and provides power to the Raspberry Pi. The motors and sensors are all controlled by the Raspberry Pi. Firmware for the board’s own microcontroller is written in Arduino, so it’s open and hackable. The hardware designs and software source code are all available online.

The case for the Raspberry Pi allows you to connect LEGO pieces to your Raspberry Pi. It protects the Pi from impacts and falls that robots tend to have, and secures all of the motors, sensors, and mechanical bits and pieces you want to attach to the Pi.

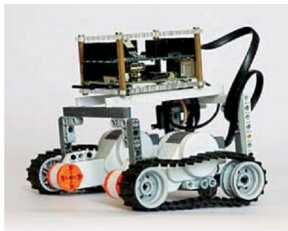
How does it work?

The device basically serves as an interface between the Raspberry Pi and LEGO Mindstorms sensors and motors, as well as a power source for the Raspberry Pi.

The BrickPi has two microcontrollers, programmed in Arduino language, each of which controls two motors and reads the motor encoder values and two sensor values. The microcontrollers are controlled and polled by the Raspberry Pi over serial lines.

“We chose Arduino platform because it is well understood and we wanted to encourage folks to improve and tweak the firmware the device runs,” says Cole.

The Arduinos can



BrickPi as a tank